MILITARY SPECIFICATION

COMPUTER-INDICATOR (SCALER), RADIAC, CP-792( )/UD

1. SCOPE

1.1 This specification covers a militarized Radiac-Computer-Indicator (Scaler) equipment for ship and shore use. The equipment is for use in computing and indicating the total number of random pulses received from external radio-activity detecting instruments, and for supplying energizing voltage to such instruments.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-Q-9858	-	Quality Program Requirements
MIL-F-15733	-	Filters, Radio Interference, General Specification for
MIL-E-16400	-	Electronic Equipment, Naval Ship and Shore, General Specification (Navy)
MIL-E-17555	-	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-P-24052	-	Preliminary Plans and Test Procedure Requirements for Radiac Instruments

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108	-	Definition of and Basic Requirement for Enclosure for Electric and Electronic Equipment
MIL-STD-167	-	Mechanical Vibration of Shipboard Equipment
MIL-STD-189	-	Racks, Electrical Equipment, 19 inch and Associated Panels
MIL-STD-415	-	Test Provisions for Electronic Systems and Associated Equipment, Design Criteria for
MIL-STD-470	-	Maintainability Program Requirements (for Systems and Equipments)
MIL-STD-471	-	Maintainability Demonstration
MIL-STD-781	-	Reliability Tests Exponential Distribution
MIL-STD-785	-	Reliability Program for Systems and Equipment Development and Production
MIL-STD-1472	-	Human Engineering Design Criteria for Military Systems, Equipment and Facilities

HANDBOOKS

MILITARY

MIL-HDBK-217	-	Reliability Stress and Failure Rate Data for Electronic Equipment
MIL-HDBK-472	-	Maintainability Prediction

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2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd. St., New York 16, N. Y.)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the command or agency concerned or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article sample. Prior to beginning production a sample shall be tested as specified in 4.3 (see 6.3).

3.2 General requirements. The equipment shall be in accordance with all provisions of MIL-E-16400, except as hereinafter specified.

3.2.1 Solid state circuitry shall be employed unless otherwise approved by the command or agency concerned.

3.3 Reliability.

3.3.1 Quantitative reliability requirement. The Radiac-Computer-Indicator (Scaler) equipment shall have a specified mean-time-between-failure of 2000 hours.

3.3.2 Reliability program plan. The contractor shall prepare for approval a reliability program plan in accordance with the requirements of this procurement and the requirements of MIL-STD-785. The reliability program plan shall describe the tasks to be performed and the procedures for conducting and controlling the reliability program. A detailed reliability program plan shall be submitted to the command or agency concerned thirty (30) days after contract award, for Government approval (see 6.2).

3.3.3 Reliability prediction. A design reliability prediction shall be performed in accordance with MIL-HDBK-217. The prediction shall be based on the operational requirements of this specification. The reliability mathematical model and reliability block diagram shall be submitted with the reliability prediction report. Mechanical parts shall be included in the prediction. The source and applicability of all failure data used for the prediction shall be documented and justified prior to use. The results of the prediction shall be compared to the specified MTBF and where the predicted values do not achieve the requirements, the contractor shall accomplish such changes in design as are necessary to increase the predicted values sufficiently to achieve the requirements. An initial reliability prediction report shall be submitted 150 days after contract award (see 6.2).

3.3.4 Reliability reports. The contractor shall prepare a report on the procedures, results and conclusions for each of the reliability tasks. Reports shall be submitted within ten (10) days after the completion of each task of the reliability program to the command or agency concerned (see 6.2).

3.4 Design requirements.

3.4.1 Component parts. The equipment shall consist of the following component parts required to make up a complete equipment:

(a) One computer-indicator (scaler).
(b) One high-voltage cable 3 feet long.
(c) One power cable 6 feet long.
(d) One technical manual.

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3.4.1.1 Computer-indicator (scaler). The equipment shall include input circuitry, amplifier, discriminator, scaling stages, stable oscillator, count indicators, as necessary to count random-pulses within the limits specified herein; plus a high voltage power supply to energize external detectors.

3.4.1.1.1 Primary power. The equipment shall be designed to operate from a supply line of 115 volts, alternating current (a.c.), 60 Hertz (Hz) per second.

3.4.1.1.2 Accuracy. The counter accuracy shall be ± one count of the total number of counts accepted by the scaler.

3.4.1.2 Discriminator adjust control. The discriminator adjust control shall be designed to permit negative pulses of 0.2 to 20 volts peak amplitude to be selected, thereby allowing all pulses between the value selected and 65 volts peak amplitude to be counted on a one-for-one basis. High level input pulses shall not produce double pulses or no pulses. This control shall be a ten turn precision potentiometer with dual dials to indicate the relative position of its contact arm; the outer dial shall show hundreds of units from 1 through 10, and the inner dial shall show units up through 99. Together the dials shall provide 1000 adjustment index points.

3.4.1.3 Input impedance. The input impedance shall be one megohm over the frequency range 50 Hz to 5 MHz.

3.4.1.4 Test signal. The equipment shall include a circuit to furnish a pulsed test signal at 100 Hz to actuate the scaling circuit. The unshaped signal shall be obtained from the stable internal frequency source used to drive the timer (see 3.4.1.10.2). This test signal shall be applied to the input of the instrument (including the input amplifier and discriminator).

3.4.1.5 Signal resolution. The equipment shall be capable of resolving individual pulses separated by not more than 1 microsecond.

3.4.1.6 Noise rejection. The equipment shall reject all pulses of less than 0.2 volts peak amplitude.

3.4.1.7 Power line isolation. Power line conducted (external noise) shall be sufficiently suppressed to prevent interference with the count. Both legs of the power line shall be filtered with a filter as specified in MIL-F-15733. The filter shall have an insertion loss of 50 decibels (db) or greater for frequencies between .15 MHz and 40 NHz. The power transformer shall contain an electrostatic shield between the primary and secondary windings with said shield grounded.

3.4.1.8 Scaler. The scaler shall consist of scaling stages and such associated circuitry as required to furnish decimal counts to the count indicator.

3.4.1.9 Count indicator. The count indicator shall consist of six decades, capable of accumulating and presenting a count up to 999,999 in the decimal system. The indicating shall be accomplished with nixie tubes. The nixie tubes display shall be capable of being dimmed electronically.

3.4.1.10 Timer. A timing system shall be provided to measure and indicate the time during which a count is made. The basic time reference shall be the internal frequency source. Time shall be indicated in decimal units of minutes up to 999.99 minutes with error in the overall indication of not more than ±1 unit in the least significant decimal place.

3.4.1.10.1 Timing controls. Timing controls shall be as follows:

Preset timer. A control shall be provided to preset the intended counting time, so that the counter and timer will stop automatically and simultaneously at the end of the preset time. Preset time shall be adjustable to the maximum timing indication of 3.4.1.10. The count and time indicators shall retain their indications after a counting run has been stopped until the indicators are reset, or the equipment turned off. It shall not be possible to start a new count without resetting time and count indicators to zero.

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3.4.1.10.2 Internal frequency source. An internal stable oscillator shall control the test signal and timer. The oscillator shall be accurate to at least 1 part in 10^4 sec.

3.4.2 High voltage power supply. A high voltage power supply shall be included, to provide voltage for the operation of external detectors.

3.4.2.1 Voltage range. The voltage range shall be continuously adjustable from plus 500 volts to plus 2500 volts.

3.4.2.2 Stability. Variation shall be not more than plus or minus 1/2 percent (±0.5%) at all voltage settings, after warmup, over a period of 1 hour.

3.4.2.3 Regulation. Regulation shall be as follows:

(a) Line. Not more than one percent (1%) for voltage change of 10 percent or frequency change of 5 percent, in primary input power.
(b) Load. Not more than one percent (1%) for variation in load between zero and 1/4 milliampere (0-0.25 ma) over the full range.

3.4.2.4 Power supply control. Control shall be by means of a ten-turn calibrated potentiometer and high-voltage switch.

3.4.2.5 Indication. The high voltage indication shall include the following:

(a) A pilot light, which shall be ON when the high voltage switch is ON.
(b) An indicating meter with scale length of 2.5 inches. The meter scale shall read from zero to 2500 volts in 50 volt scale division.
(c) The 2500 volt panel meter shall indicate the high voltage with an error of ±2 percent or less of full scale.

3.4.2.6 Short-circuit protection. Short-circuit protection shall be provided for the high voltage supply.

3.4.3 Control panels. The controls, indicators, and connectors shall be located as follows:

(a) Front panel:
(1) Switch, master.
(2) Pilot light, master.
(3) Switch, count and time, manual.
(4) Switch, preset time selector.
(5) Indicator, counts accumulated.
(6) Indicator, time of count.
(7) Switch, high voltage.
(8) Pilot light, high voltage.
(9) Control knob, high voltage.
(10) Indicating meter, high voltage.
(11) Discriminator calibration adjust.

(b) Rear panel:
(1) Input sensitivity adjust.
(2) Adjustments for internal voltages.
(3) Connection, primary power.
(4) Connector, counter input, with high voltage. Connector shall be rated at 2500 volts.
(5) Fuses, spare fuses, and blown fuse indicators.
(6) Test points, of the type designated "accessible" in accordance with MIL-STD-415.
(7) Power for preamplifier.
(8) Connector for preamplifier. Connector shall be rated at 2500 volts.

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3.4.3.1 Locking devices. Controls on the front panel shall not have locking devices. Controls on the rear panel shall have locking devices.

3.4.3.2 Elapsed time meters. No timers other than the timer specified in 3.4.1.10 shall be furnished.

3.4.4 Cabinet. The equipment shall be housed in a cabinet of minimum size and weight consistent with other requirements. Shock and vibration resistance shall be attained without the use of shock mounts. The degree of enclosure shall be "dripproof" in accordance with MIL-STD-108. The form factor and arrangement of controls, connectors, and indicators shall be subject to the approval of the command or agency concerned.

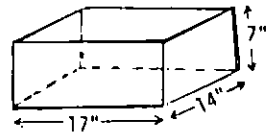
3.4.4.1 Chassis. The chassis shall be such that it can be removed from the cabinet and mounted in a standard 19 inch electrical equipment rack in accordance with MIL-STD-189. The chassis shall be on slides to facilitate removal from the cabinet housing. The chassis tracks shall enable the chassis to be extended from the cabinet and rotated through 105 degrees for easy servicing. The safety requirements of MIL-E-16400 shall be observed.

3.4.5 Decontamination. Parts subject to contamination shall be readily capable of decontamination without recourse to laboratory methods or techniques. The housing shall have a smooth surface and present a minimum of crevices and sharp edges. Rubber or felt feet will not be permitted on the case. All indication marking scales engraved on the case shall be filled with a nonporous paint of contrasting color to produce a plainly marked, smooth surface.

3.4.6 Technical manual. The technical manual shall be as specified in applicable schedule.

3.4.7 Environmental conditions. The equipment shall meet the temperature requirements of class 4 and the humidity requirements of MIL-E-16400.

3.4.8 Size and weight. The equipment shall be rectangular with maximum dimensions as follows:



The weight shall not exceed 50 pounds.

3.4.9 Preliminary plans. Preliminary plans shall be submitted in accordance with MIL-P-24052 (see 6.2).

3.5 Mechanical.

3.5.1 Shock, vibration, inclination and noise. Inclination and noise do not apply to this specification. The shock requirements shall be for grade A, class 1, lightweight. The vibration requirements shall be type 1 of MIL-STD-167.

3.6 Maintainability.

3.6.1 Quantitative maintainability requirements. The Radiac-Computer-Indicator (Scaler) shall have an equipment repair time (ERT) of not greater than thirty (30) minutes.

3.6.2 Maintainability program plan. The contractor shall prepare for approval a maintainability program plan in accordance with the requirements of this procurement and the requirements of MIL-STD-470. The maintainability program plan shall describe the tasks to be performed and the procedures for conducting

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and controlling the maintainability program. A detailed maintainability program plan shall be submitted to the command or agency concerned thirty days after contract award, for Government approval (see 6.2).

3.6.3 Maintainability prediction. A maintainability prediction shall be performed using procedure II of MIL-HDBK-472 The contractor shall document and justify all assumptions and the applicability of all data used in the prediction. The prediction shall also include mechanical parts. The results of the prediction shall be compared to the required ERT and where the predicted values do not achieve the requirements the contractor shall accomplish such changes in design as are necessary to improve the predicted values sufficiently to achieve the requirements. An initial maintainability prediction report shall be submitted 180 days after contract award (see 6.2).

3.6.4 Maintainability reports. The contractor shall prepare a report on the procedures, results and conclusions for each of the maintainability tasks. Reports shall be submitted within ten (10) days after the completion of each task of the maintainability program to the command or agency concerned (see 6.2).

3.7 Human factors. The contractor shall apply the human engineering principles of MIL-STD-1472 to the Radiac-Computer-Indicator (Scaler). The panel design shall be submitted to the command or agency concerned for review and approval prior to equipment production.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality program requirement. The contractor shall provide and maintain a quality program acceptable to the Government for supplies and services covered by this specification. The quality program shall be in accordance with MIL-Q-9858.

4.1.1.1 Quality plan. A quality plan, applicable to the requirements of this specification shall be submitted to the Quality Assurance Representative for approval prior to first article inspection. If first article inspection is not required the quality plan shall be submitted for approval prior to quality conformance inspection. The quality plan shall reflect the following (see 6.2):

(a) flow of material from inspection on receipt through all manufacturing processes to final shipment.
(b) process and inspection points.
(c) reference to process and inspection procedures.
(d) differentiation between in-plant inspections and subcontracted inspections for subassemblies and higher assemblies.

Any changes to the approved plan shall be submitted to the Quality Assurance Representative for approval prior to use.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all the requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

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4.1.3 Tests, test equipment and test procedures. The contractor shall submit written test procedures in accordance with MIL-P-24052 and as specified herein. Test procedures shall include test equipment and tolerance limits to be used for first article inspection and quality conformance inspection to the command or agency concerned, via the Government inspector for review and approval prior to beginning tests (see 4.2 and 6.2).

4.1.3.1 Procedures for inspections, tests, demonstrations, and so forth, shall contain step by step information which will enable the operator to properly perform the requirements of this specification. The method of examination shall be clearly defined.

4.1.3.2 The procedures shall contain a list of test equipment identified by name and model, range and accuracy. Special test equipment shall also be listed and identified.

4.1.3.3 Data sheets shall be included in the procedures. Conformance parameters shall be clearly indicated with tolerance limits. Whenever quantitative measurements are made, the result shall be shown by quantitative data.

4.2 Classification of inspection. The method of examination and testing of the equipment fall within the following classifications:

(a) First article inspection

(b) Quality conformance inspection
  (1) Production inspection
  (2) Production control inspection
  (3) Environmental inspection
  (4) Inspection of preparation for delivery

(c) Reliability testing

(d) Maintainability demonstration

4.3 Unless otherwise specified (see 6.1), four units shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in table I.

TABLE I. First article inspection

Examination or test	Requirement paragraph	Test method paragraph	Number of sample units to be inspected	Number of failures
Surface examination	3.2, 3.4.1, 3.4.3, 3.4.4, 3.4.5, 3.4.6	4.5	4	0
Operating test	3.4.1.1.2, 3.4.1.3, 3.4.1.5, 3.4.1.6, 3.4.1.9, 3.4.1.10, 3.4.2.1, 3.4.2.2, 3.4.2.5(a) and (c).	4.5	4	0
Weights and dimensions	3.4.8	4.5	4	0
Supply line voltage and frequency	3.4.1.1.1	4.5	4	0
Heat test	3.2	4.5	4	0
Enclosure test 1/	3.4.4	4.5	4	0
Temperature	3.4.7	4.5	4	0
Humidity	3.4.7	4.5	4	0
Power	3.2	4.5	4	0

1/ Enclosure test shall be conducted after completion of shock/vibration

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TABLE I. First article inspection - Continued

Examination or test	Requirement paragraph	Test method paragraph	Number of sample units to be inspected	Number of failures
Controls and control circuits	3.4.1.2, 3.4.1.10.1, 3.4.1.10.2, 3.4.2.3, 3.4.2.4, 3.4.3(a)(1), 3.4.3(b)(2), 3.4.3(b)(3), 3.4.3(b)(6)	4.5	4	
Shock	3.5.1	4.5	4	
Vibration and structural integrity	3.5.1	4.5		
Radio frequency interference emission and susceptibility test	3.4.1.7	4.5	4	
Salt spray test	3.2	4.5	4	

4.3.1 First article inspection report. A First article inspection report containing the results of all tests shall be submitted to the command or agency concerned for approval upon completion of First article inspection. The test report for structural integrity, however, shall be submitted within the 20-day period specified in MIL-E-16400. Report of shock tests shall include completed Form NAVEXOS 3373. Report demonstrating ability of the equipment to perform its intended function during and after shock and vibration tests shall be included. First article approval will not be granted unless the report is complete. Any production entered into prior to approval of the First article report will be at the supplier's risk (see 6.2).

4.4 Quality Conformance Inspection.

4.4.1 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall consist of the procedures specified in 4.1.3 and shall include the requirements of table II.

TABLE II. Production Inspection

Examination of test	Requirement paragraph	Test method paragraph
Surface examination	3.2, 3.4.1, 3.4.3, 3.4.4, 3.4.5, 3.4.6	4.5
Operating test	3.4.1.1.2, 3.4.1.3, 3.4.1.4, 3.4.1.5, 3.4.1.6, 3.4.1.9, 3.4.1.10, 3.4.2.1, 3.4.2.2, 3.4.2.5(a), 3.4.2.5(c)	4.5

4.4.2 Production control inspection. Production control inspection, including sampling, shall conform to table III and to the procedures for special inspection levels of MIL-STD-105. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on equipments that have passed production inspection. Production control inspection shall be approved in accordance with 4.1.3. The equipment shall satisfactorily meet the requirements of production control inspection prior to release for shipment.

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TABLE III. Production control inspection

Inspection	Requirement paragraph	Inspection paragraph	AQL (percent defective)
Weights and dimensions	3.4.8	4.5	10
Supply line voltage and frequency	3.4.1.1.1	4.5	6.5
Heat Test	3.2	4.5	10
Enclosure test	3.4.4	4.5	6.5
Power	3.2	4.5	6.5
Controls and Control Circuits	3.4.1.2, 3.4.1.10.1, 3.4.1.10.2, 3.4.2.3, 3.4.2.4, 3.4.3(a)(1), 3.4.3(b)(2), 3.4.3(b)(3), 3.4.3(b)(6)	4.5	6.5

4.4.2.1 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots.

4.4.3 Environmental inspection. Environmental inspection shall be conducted to prove the durability of the equipment to withstand the effects of extremes of environment, service conditions, and hard usage. Environmental inspection shall be performed on equipments that have passed production inspection and shall include the tests listed in table IV.

TABLE IV. Environmental inspection

Examination of test	Requirement paragraph	Test method paragraph
Temperature	3.4.7	4.5
Humidity	3.4.7	4.5
Shock	3.5.1	4.5
Vibration	3.5.1	4.5

4.4.3.1 Environmental inspection samples. Samples for the environmental inspection shall be selected without regard to their quality in accordance with the following:

(a) One sample from the first 25 production units fabricated.
(b) One sample from each successive 100 units produced, or fraction thereof.

4.4.4 Reinspection of conforming production control and environmental inspection sample units. Unless otherwise specified, sample units which have been subjected to and passed production control and environmental inspections may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all visible damage.

4.4.5 Reliability testing. The reliability of the Radiac-Computer-Indicator (Scaler) shall be demonstrated and reported in accordance with the requirements of MIL-STD-781.

4.4.5.1 Failure. Failure shall be defined as any malfunction or parameter deviation that prevents the equipment from performing within the operational requirements set forth in this specification. All failures shall be considered relevant unless determined by the command or agency concerned to be caused by a condition external to the equipment under test which is not a test requirement and not encountered in service.

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4.4.5.2 Reliability test plan. The contractor shall prepare a detailed reliability test plan in accordance with MIL-STD-781. Test parameters and conditions shall be in accordance with the equipment operational requirements as defined in the equipment specification. Performance parameters to be measured during the reliability test shall be in accordance with the equipment specification. Debugging and burn-in prior to the reliability test shall be limited to the debugging and burn-in to be performed on all production equipment prior to delivery to the Government. No preventative maintenance shall be performed during the reliability test. At least ninety (90) days prior to the start of reliability testing, the contractor shall submit a reliability test plan to the command or agency concerned for Government approval (see 6.2).

4.4.5.3 Performance parameters for reliability testing. The parameters to be monitored shall include the following:

(a) Input sensitivity (see 3.4.1.2)
(b) Test signal (see 3.4.1.4)
(c) Signal resolution (see 3.4.1.5)
(d) Noise rejection (see 3.4.1.6)
(e) Power line isolation (see 3.4.1.7)
(f) Count indication (see 3.4.1.9)
(g) Timer operation (see 3.4.1.10)
(h) Timing controls (see 3.4.1.10.1)
(i) High voltage supply (see 3.4.2)
(j) Short circuit protection (see 3.2.6)

4.4.5.4 Reliability qualification tests. Reliability qualification testing shall be conducted on the initial production units. Testing will be performed at Test Level C with Test Plan IV. Qualification testing shall be performed on four (4) production units Reliability testing shall be successfully completed prior to equipment shipment.

4.4.5.5 Reliability sampling tests. Reliability sampling tests shall be conducted throughout production to insure continued production of acceptable equipment. Sampling testing will be performed at Test Level C with Test Plan IV A. MIL-STD-781 shall be used to determine the number of units to be selected for reliability sampling tests. Reliability testing for each lot shall be successfully completed prior to equipment shipment.

4.4.5.6 Final restoration and disposition of tested equipment. Upon completion of reliability testing, the tested equipment shall be returned to production for reinspection and reconditioning as required, and may be delivered as part of the contract quantity if it meets all terms and conditions of the contract for acceptance.

4.4.5.7 Reliability test reports. The contractor shall submit test reports for reliability testing in accordance with MIL-STD-781. Reliability test reports shall be submitted within ten days after completion of each reliability test to the command or agency concerned for Government approval (see 6.2).

4.4.6 Non-conformance and retest. If a sample unit fails a test specified in 4.4.2, 4.4.3 or 4.4.5, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken to correct units of product which were manufactured with the same conditions, materials, processes, and so forth. The equipment shall then be retested.

4.4.7 Corrective action. If corrective action results in a change of part(s), circuit layout or mechanical arrangement, or if the government inspector does not consider that corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred immediately to the contracting officer. It shall then be determined by the command or agency concerned whether correction should be made to the equipment already delivered under the contract.

4.4.8 Maintainability demonstration test. The maintainability requirements for the equipment shall be successfully demonstrated in accordance with method III of MIL-STD-471. The maintainability test shall be performed on the first article.

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4.4.8.1 Maintainability demonstration test plan. The contractor shall prepare a detailed maintainability test plan in accordance with MIL-STD-471. Ninety (90) days prior to the start of testing, the contractor shall submit the maintainability demonstration test to the command or agency concerned for Government approval (see 6.2).

4.4.8.2 Maintainability demonstration report. A final maintainability demonstration report shall be submitted in accordance with MIL-STD-471. The maintainability demonstration report shall be submitted to the command or agency concerned for Government approval (see 6.2).

4.4.9 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation and packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

4.5 Inspection procedures. Inspection procedures shall be in accordance with MIL-E-16400.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation and packaging, packing, and marking. Unless otherwise specified, preparation for delivery shall be in accordance with MIL-E-17555 (see 6.1).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of the specification.
(b) Requirement for technical manual (see 3.4.6)
(c) Number of first article samples to be submitted if other than specified in 4.3.
(d) Marking, levels of preservation and packaging and packing required (see 5.1)

6.2 Data generated by this specification. Data generated by this document is not deliverable unless specified on the Contract Data Requirements List (DD Form 1423). The data required by this specification includes, but is not restricted to the following:

(a) Reliability program plan (see 3.3.2).
(b) Reliability prediction report (see 3.3.3).
(c) Reliability reports (see 3.3.4).
(d) Preliminary plans (see 3.4.9).
(e) Maintainability program plan (see 3.6.2).
(f) Maintainability prediction report (see 3.6.3)
(g) Maintainability report (see 3.6.4)
(h) Quality plan (see 4.1.1.1)
(i) Test procedures and data sheets (see 4.1.3)
(j) First article inspection report (see 4.3.1)
(k) Reliability test plan (see 4.4.5.2)
(l) Reliability test reports (see 4.4.5.7)
(m) Maintainability demonstration test plan (see 4.4.8.1)
(n) Maintainability demonstration report (see 4.4.8.2)

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6.3 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy - EC

(Project 6665-N260)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter)

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)

DATE

DD FORM 1 APR 63 1426

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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